

Performance-Focused Design Exercise Packet

Part of ASME IDETC 2025 Workshop:

Adventures in Tradespace Exploration

17 August 2025

Group ID: 4025



Annotate, mark-up, and write on this page as needed.

Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables.

very Narrow	Long	Modest High roof							
102 in	923 in	96.1 in	8.6 in	715 lb-ft	479.1	45 in	990 $\frac{\text{kw}}{\text{hr}}$	0.73 in	0.28 in

Do you think you could have found a better solution given more time?

YES

What, in this round, stopped you from finding a better solution?

Time constraint, Lack of data,

DO NOT MOVE TO NEXT PAGE UNTIL INSTRUCTED

Annotate, mark-up, and write on this page as needed.

Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables.

110 in	80.1 in	65.5 in	20.8 in	421 lb-ft	2:15:1	49 in	1660 kWh	0.939 in	1.48 in
--------	---------	---------	---------	-----------	--------	-------	----------	----------	---------

Do you think you could have found a better solution given more time?

YES

What, in this round, stopped you from finding a better solution?

Even though we found an out & out winner, the decision process ~~would~~ could've been even better if we were provided with more information like the performance specs of each design ID.

DO NOT MOVE TO NEXT PAGE UNTIL INSTRUCTED

Post-Activity Questions:

Describe the solution the group has selected – write in the value of the design variables

181 in	81.2 in	69.2 in	9.7 in	655 lb-ft	2:37:1	32 in	1070 kWh	0.877 in	1.99 in
--------	---------	---------	--------	-----------	--------	-------	----------	----------	---------

Do you think you could have found a better solution given more time?

Not in this round. Most of the parameters we ~~to~~ were looking for were favourable in Design #4 hence why we ~~ended~~ think that we couldn't have found a better solution.

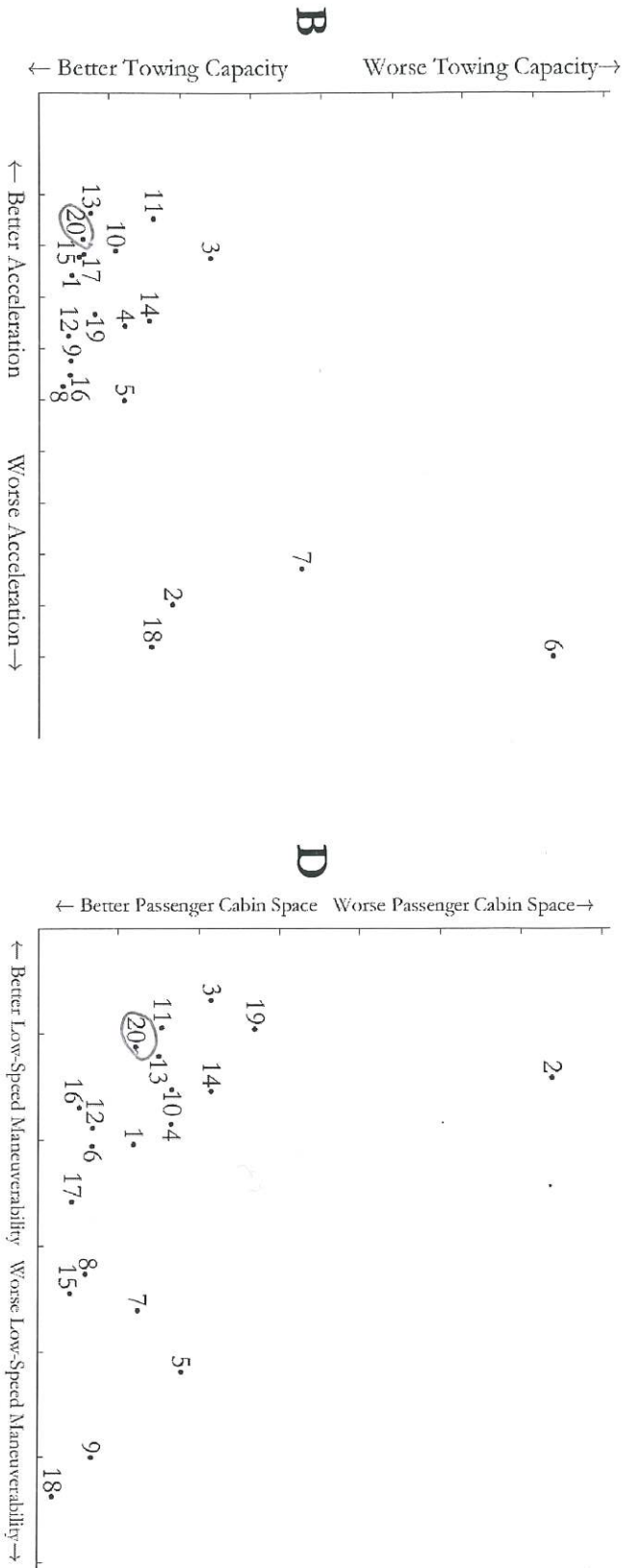
What, in this round, stopped you from finding a better solution?

DO NOT MOVE TO NEXT PAGE UNTIL INSTRUCTED

Vehicle Index

	TW	WB	RH	ST	MET	DG	TD	EC	FRWT	BPT
15	200 in	95.1 in	81.7 in	20.6 in	665 lb-ft	5.92:1	31 in	1460 kWh	0.313 in	1.67 in
	Very Wide	Long	Very Tall Roof	Very Long Travel	High Output	Very Short Gearing	Very Small Tire	High Energy Capacity	Very Light-Duty Frame	Very Heavily Armored
16	124 in	99.8 in	82.1 in	18.8 in	532 lb-ft	3.62:1	31 in	1570 kWh	0.782 in	0.33 in
	Narrow	Very Long	Very Tall Roof	Long Travel	Low Output	Moderately Gearing	Very Small Tire	Very High Energy Capacity	Heavy-Duty Frame	Very Lightly Armored
17	172 in	94 in	83.4 in	9.6 in	735 lb-ft	5.94:1	43 in	1500 kWh	0.313 in	0.78 in
	Wide	Long	Very Tall Roof	Very Short Travel	Very High Output	Very Short Gearing	Large Tire	High Energy Capacity	Very Light-Duty Frame	Lightly Armored
18	192.00527	98 in	82.4 in	6.5 in	442 lb-ft	3.23:1	35 in	1120 kWh	0.308 in	1.7 in
	Very Wide	Very Long	Very Tall Roof	Very Short Travel	Very Low Output	Long Gearing	Small Tire	Low Energy Capacity	Very Light-Duty Frame	Very Heavily Armored
19	108 in	84.8 in	66.3 in	9.6 in	456 lb-ft	5.74:1	50 in	1080 kWh	0.966 in	0.84 in
	Very Narrow	Short	Very Short Roof	Very Short Travel	Very Low Output	Very Short Gearing	Very Large Tire	Low Energy Capacity	Very Heavy-Duty Frame	Lightly Armored
20	102 in	92.3 in	76.1 in	8.6 in	715 lb-ft	4.79:1	45 in	990 kWh	0.733 in	0.28 in
	Very Narrow	Long	Moderate Height Roof	Very Short Travel	Very High Output	Short Gearing	Large Tire	Very Low Energy Capacity	Heavy-Duty Frame	Very Lightly Armored

Annotate, mark-up, and write on this page as needed.

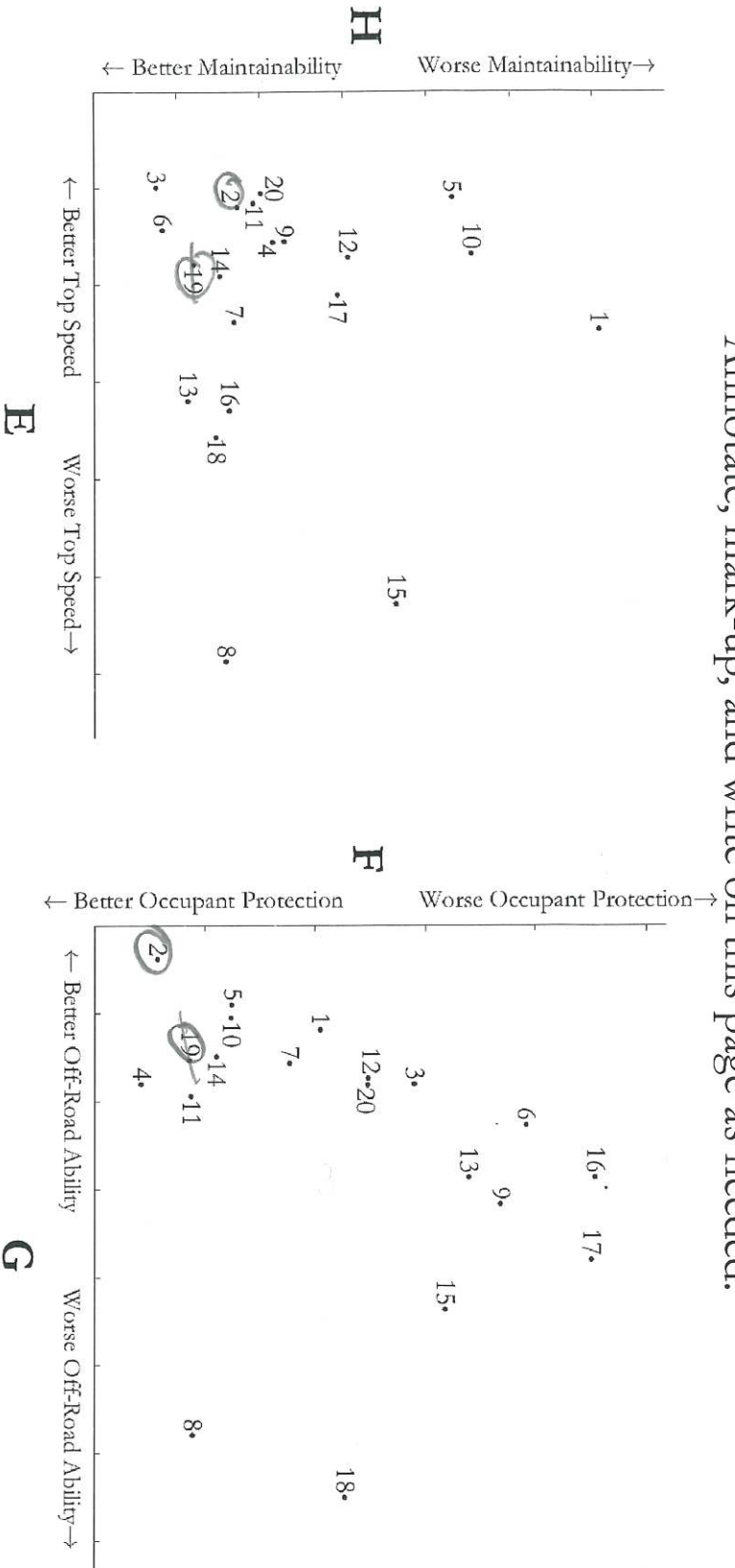


It came down to #20 & #1, we ended up going with the former as it performed slightly better in the required characteristics.

Enter the number ID of the selected design: 20

Write your Group ID: 4025

Annotate, mark-up, and write on this page as needed.

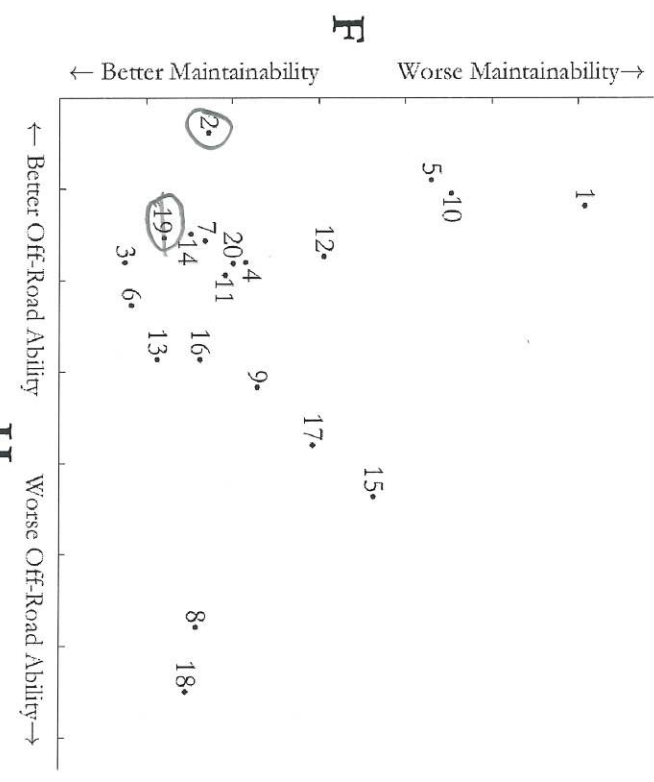
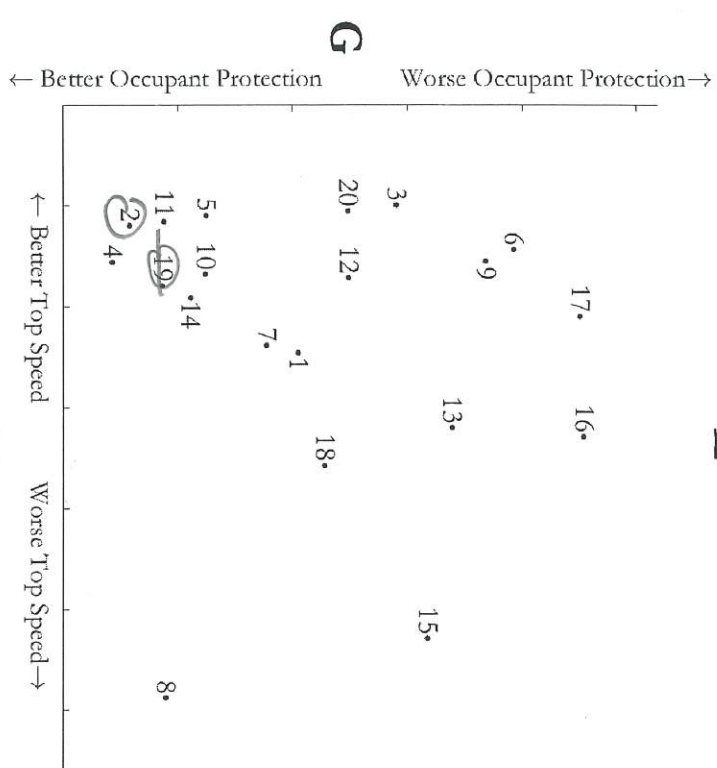
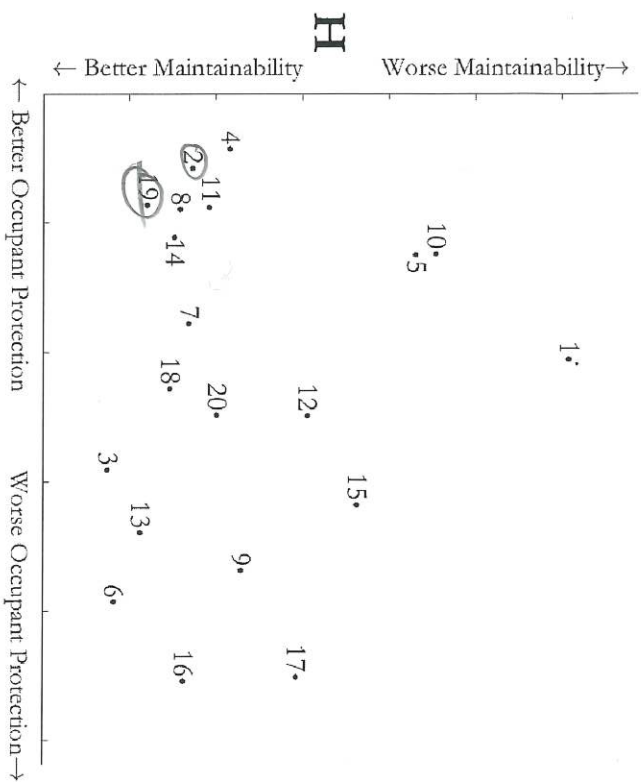
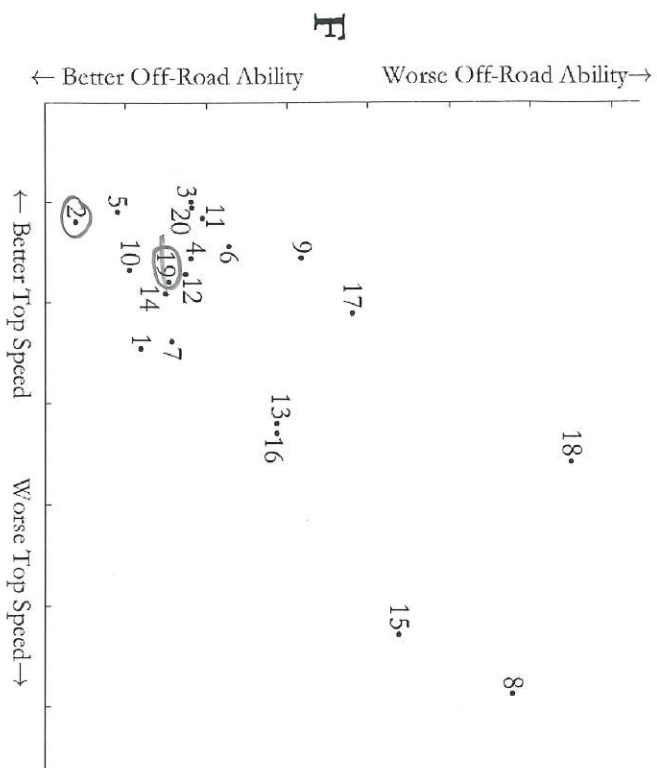


It came down to #2 & #19, we ended up going with the former as #2 has better off road ability, occupant protection, top speed as compared to #19

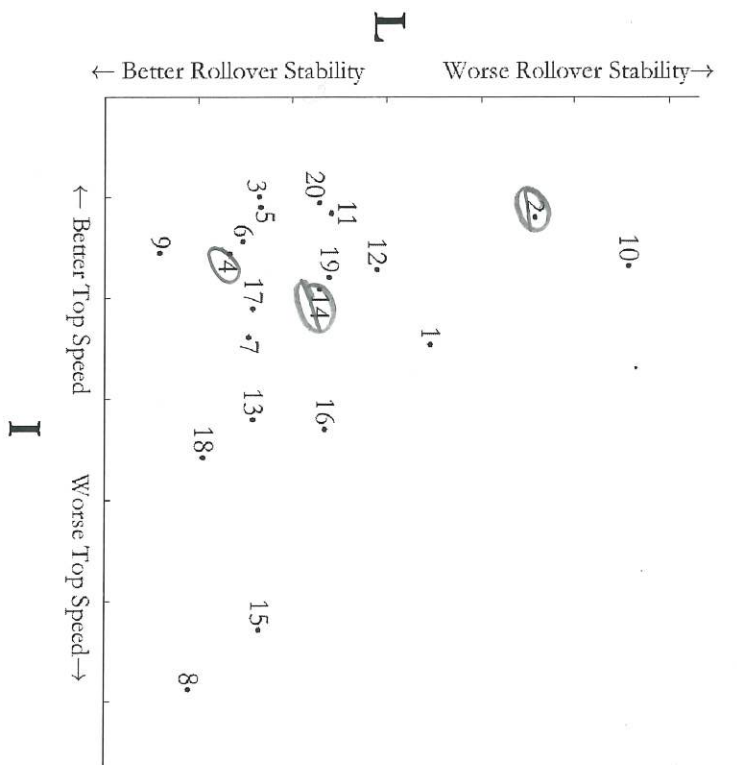
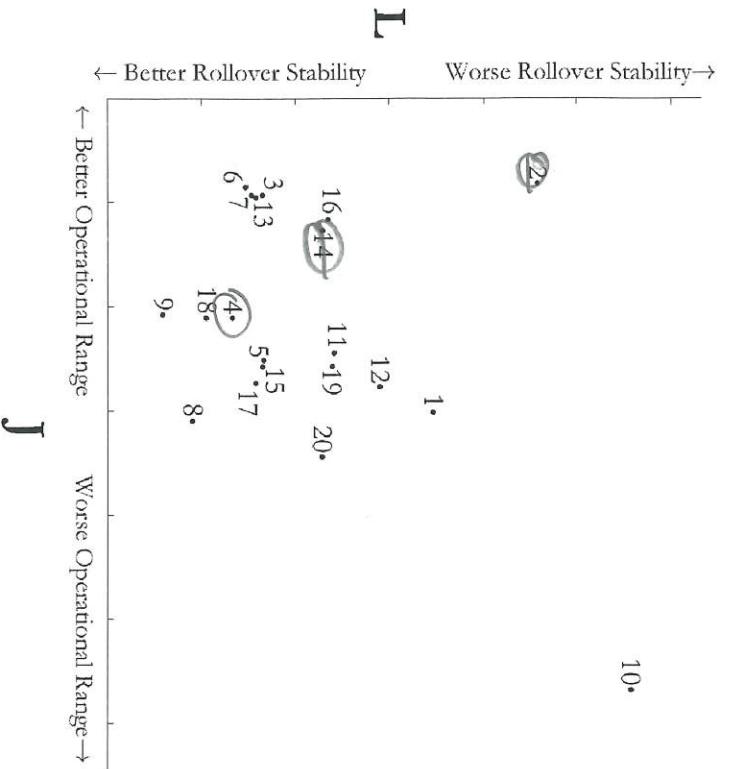
Enter the number ID of the selected design: 2

Write your Group ID: 4025

Annotate, mark-up, and write on this page as needed.



Annotate, mark-up, and write on this page as needed.



It came down to # 2, 4, and 14, but we decided on 4. Although we compromised on Operational Range, 4 seems better in the other 3 categories when compared to 14 and 2.

Enter the number ID of the selected design: 4

Write your Group ID: 4025

Annotate, mark-up, and write on this page as needed.

